Mid-Range Convolutional Modulated Transformer Network for Hyperspectral Image Classification

Yuyang Wang, Mingzhu Tai, Zigao Liu, Zhenqiu Shu, *Member, IEEE*, Fengchao Xiong, *Member, IEEE*, and Zhengtao Yu

***Abstract*—Hyperspectral image (HSI) classification plays a major role in remote sensing. The combination of convolutional neural networks (CNNs) and transformers has garnered increasing attention in HSI classification, yielding impressive performance. However, these approaches primarily focus on modeling local–global feature representations, often ignoring the rich multirange information inherent in HSIs. To address these issues, this work proposes a mid-range convolutional modulated transformer network (MCMTN) for HSI classification. Compared to conventional CNN and transformer architectures, the proposed method employs large-kernel convolutions to simplify self-attention (SA) operations and facilitate the integration of short-, mid-, and long-range information in HSIs. Specifically, we employ a lightweight spatial–spectral module (LWSSM) to extract the short-range spatial features and spectral features of HSIs. Furthermore, we can effectively model the long-range spatial information in a spatial compression module (SCM) using a simple compress-decompress strategy. We also introduce a mid-range convolutional modulation (MCM) into the network, which effectively captures mid-range spatial dependencies in HSIs while enabling semantic information interaction in a transformer manner. Compared with traditional HSI classification methods, our proposed MCMTN method can effectively extract multidimensional representations with less time consumption. The experimental results on several benchmark datasets show that the proposed MCMTN method achieves better classification performance than several state-of-the-art models. The code is available at https://github.com/szq0816/MCMTN**

***Index Terms*—Convolution modulation, convolutional neural networks (CNNs), global and local spatial–spectral information, hyperspectral image (HSI) classification, mid-range, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) usually contain tens or hundreds of spectral bands, which can be represented as a 3-D data cube with multiple channels. HSIs have more abundant spectral-spatial details and discriminative features than panchromatic and color remote sensing images. Therefore, they have been used in many real-world remote sensing and Earth observation applications, such as agriculture production [1], environmental monitoring [2], and reconnaissance [3], etc. In most applications, HSI classification is an important task that aims to provide pixel-level class prediction.

With the significant progress of deep learning in various real-world problems [4], [5], [6], [7], [8], [9], [10], [11], deep learning methods have become increasingly dominant in the field of HSI classification [12], [13], [14], [15], [16]. Among them, convolutional neural networks (CNNs) are the most representative method due to their excellent feature extraction and image representation capability. Hu et al. [17] proposed to extract discriminative spectral features of HSIs using 1-D convolution, and subsequently obtained land-cover classification using two feed-forward neural layers. After that, many efforts [18], [19] focused on utilizing both spectral information and spatial information simultaneously. Yang et al. [18] used 2-D convolutional and 1-D convolutional to extract spatial and spectral information of HSIs, respectively. Then the spatial–spectral features were fed into a fully connected layer to get the classification results. Furthermore, 3-D convolution [19] was developed to extract the spatial–spectral features in a single convolution operation.

Despite the great success in HSI classification, CNNs have limited capability in long-range context modeling. To solve this problem, transformers have been proposed to enable the long-distance modeling capability of deep neural networks [20], [21]. Hong et al. [22] proposed a SpectralFormer approach by applying the vision transformer (ViT) to HSI classification tasks, in which group-wise spectral embedding is used to add locality into long-distance spectral dependencies. Yang et al. [23] used 3-D convolution instead of a linear projection layer in the transformer's encoder, which can more effectively capture subtle spectral-spatial discrepancies. Zhong et al. [24] designed a novel factorized architecture search framework to determine the layer-level operation choices and block-level orders of both spectral and spatial transformer blocks.

CNNs and transformers have their unique pros and cons. First, CNNs mainly focus on relative spatial locations between pixels rather than absolute spatial locations [25]. This translation equivalence of the convolution property significantly enhances the model's generalization capabilities on small-scale datasets [26]. However, transformer-based HSI classification methods do not have this property, so they

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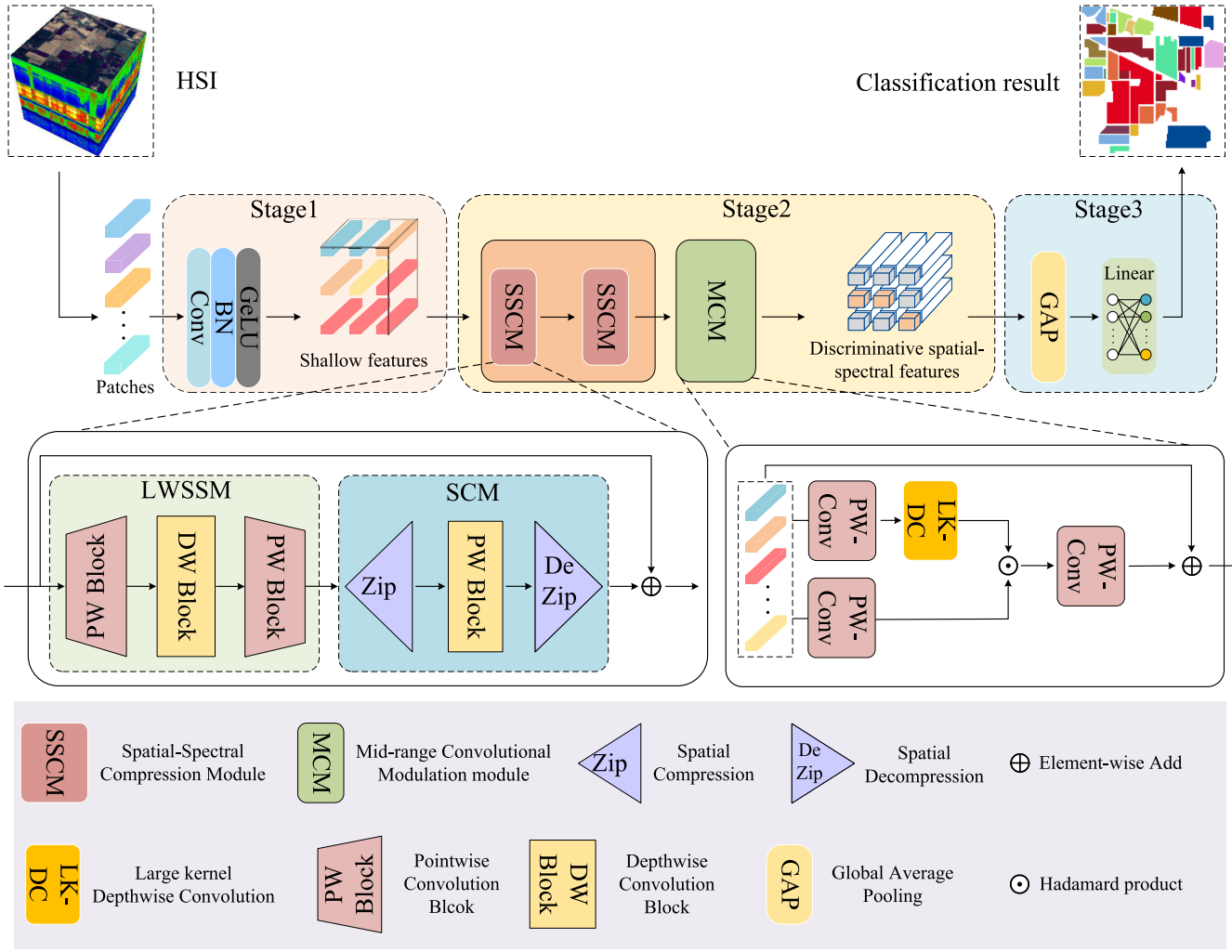


Fig. 1. Framework structure of the proposed MCMTN method. It is composed of three stages: initial feature extraction, triple-range feature extraction, and classification. The initial feature extraction stage feeds the extracted low-level semantic features to the triple-range feature extraction stage, which further captures multigranularity spectral–spatial information. The fused features are fed into the classification stage for final prediction.

cannot effectively deal with small-scale datasets [27]. Second, the weights of the convolutional kernel are usually fixed for the input, whereas the attention weights in transformers are generated adaptively based on the input [28]. As a result, transformer networks can capture token dependencies dynamically with great flexibility. Third, self-attention (SA) and convolution have significant differences in the size of their receptive fields. Generally speaking, a larger receptive field provides more contextual information. Moreover, the global receptive field is considered to be an important characteristic of the transformer. Therefore, in recent years, several studies [29], [30], [31] have fully utilized the respective advantages of CNNs and transformers in HSI classification. Shi et al. [31] utilized local features extracted from CNN layers as input tokens to the transformer, alleviating the insufficient local perception ability of the transformer. Ding et al. [29] and Zhou et al. [30] fused features from CNN and transformer branches to obtain more discriminative local–global representations. However, they cannot effectively explore the multirange spatial dependence of HSIs, which limits their ability to learn comprehensive representations. Xu et al. [32] introduced multiscale convolutional layers before the transformer, but this approach fails to fully overcome the transformer's limited ability to perceive mid-range details. Wu et al. [33] employed multiscale patch inputs, which inevitably increases the computational overhead. Efficiently utilizing the multirange information inherent in HSIs remains a highly challenging topic in HSI classification.

In this article, we propose a novel end-to-end learning framework, called mid-range convolutional modulated transformer network (MCMTN), for HSI classification. The proposed MCMTN method aims to capture dependencies across three different spatial ranges in HSIs and thus extract richer discrimination features. The overall architecture of the proposed method is shown in Fig. 1. First, we design a spatial–spectral compression module (SSCM) to extract both different short- and long-range spatial and spectral information from HSIs. To capture mid-range spatial dependencies, we design a mid-range convolutional modulation (MCM) module. It employs a large-kernel convolutional modulation operation

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to simulate the SA mechanism in transformers. Finally, the spatial–spectral discriminative features are fed into a fully connected layer, and then the label of each pixel can be accurately predicted. Extensive experimental results on several benchmark datasets show that the proposed method outperforms other state-of-the-art HSI classification approaches.

The main contributions of this work can be summarized as follows.

1) We propose an end-to-end convolution-transformer network to extract multirange spatial and spectral features for HSI classification. Compared with traditional HSI classification methods, the proposed framework fully leverages the strengths of convolution and transformer, thus effectively extracting high-level semantic information from HSIs.
2) We develop the SSCM to efficiently extract both short- and long-range spatial and spectral information from HSIs. Specifically, it consists of the newly designed lightweight spatial–spectral module (LWSSM) and spatial compression module (SCM). The LWSSM performs lightweight spectral and short-range spatial feature extraction, while the SCM compresses features to effectively capture long-range spatial dependencies. The synergistic integration of these two modules not only improves overall model performance but also significantly improves the quality of feature representation in HSIs.
3) To capture the mid-range spatial dependencies between short- and long-range dependencies, we replace the original SA in ViTs with the proposed MCM module. Although our MCM module captures spatial dependencies in a transformer-like manner, it is an entirely CNN-based approach. This design enables the model to better leverage the strengths of large-kernel convolutions, thereby enhancing its capacity to capture multiscale spatial features.
4) Experimental results on three benchmark HSI datasets have shown the effectiveness of the proposed method in HSI classification.

The rest of the article is organized as follows. Section II reviews related works in HSI classification. Section III introduces the proposed approach and its optimization scheme. Section IV presents the experimental results and their analysis on different HSI benchmarks. Section V draws brief conclusions from this work.

II. Related Works

This section provides a brief review of CNN- and transformer-based HSI classification methods in remote sensing.

A. CNN-Based HSI Classification Methods

CNN is one of the most well-known HSI classification approaches due to its excellent results. To boost the classification results of HSIs, most CNN-based HSI classification approaches aim at extracting their spatial–spectral information. Roy et al. [34] proposed a hybrid spatial–spectral network. It extracts joint spatial–spectral feature representations from a series of spectral bands using 3-D-CNN and then further adds more abstract spatial representations using 2-D-CNN. However, it consumes more computational cost to perform 3-D convolution. Naturally, the training costs of the model increase dramatically when 3-D convolution is extensively used. Nikzad et al. [35] put forward a unique triangular lattice (TL) topology of 2-D convolutional kernels that provides a lightweight model with both residual and dense aggregation links. This structure can effectively reduce the computational cost by performing extensive 3-D convolution operations. The above-mentioned methods use a fixed-size receptive field and fail to capture the feature differences between different receptive fields. Alipour-Fard et al. [36] employed multiple convolution branches with different kernel sizes to extract the features of HSIs. Then they adopted a selection kernel mechanism to adaptively fuse multiple features, thereby obtaining the most discriminative features of HSIs. Roy et al. [37] proposed to dynamically adjust the network receptive field while extracting spatial–spectral joint features by designing an efficient feature recalibration mechanism. Shu et al. [38] incorporated the spatial and spectral split attention into bottleneck residual units to extract multireceptive fields across features at a finer level. Despite the strong generalization performance of CNN-based HSI classification models, they struggle to effectively model long-range spatial–spectral dependencies in HSIs.

B. ViT-Based HSI Classification Methods

In recent years, various transformer networks have been applied to the HSI classification problem [39], [40], [41], [42]. Unlike CNNs, these approaches typically require tokenizing the input samples. Specifically, the patches of HSIs are usually partitioned into nonoverlapping tokens. Each token represents the features at a specific spatial location within the HSIs. By learning the relationship between tokens, ViTs can more effectively capture long-range spatial–spectral dependencies than CNNs. However, tokenization may result in the loss of fine-grained local spatial information. To effectively preserve the spatial sequence relationships, Zou et al. [43] first used the simple linear iterative clustering (SLIC) algorithm to segment patches into superpixels. The segmentation results are then embedded into tokens to preserve local edge contour information in patches. Similarly, Tu et al. [44] proposed local neighborhood aggregation operations to merge pixel token features and learn multiscale features. Mei et al. [45] introduced a novel grouping pixel embedding (GPE) module to extract distinctive spatial–spectral features from nonoverlapping group channels using group convolution. GPE focuses on a relatively local area by combining with MHSA, effectively mitigating the over-dispersion issue in MHSA. Sun et al. [46] incorporated Gaussian distribution weights into the tokenization process, thereby transforming shallow spatial–spectral features into semantic token representations. As a result, the deep semantic token features better align with the intrinsic data distribution. Roy et al. [41] combined attention mechanisms with morphological operations to model structural and shape information, enriching the feature representations. Zhao et al. [39] proposed a hybrid encoder architecture by combining convolutions and

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transformers. It replaces traditional linear token embeddings with convolutional embeddings, thus effectively modeling local and global information. Sun et al. [47] designed a memory tokenizer (MT) to transform spectral–spatial information into memory tokens, and then employed a memory-augmented transformer encoder (MATE) to effectively extract joint features. Ma et al. [48] proposed a lightweight Gaussian SA to focus on the classification target by enhancing the feature weight of the central pixel. Chen et al. [49] utilized center rotation and relative position embedding to reduce the impact of absolute directional information on classification. Furthermore, they designed a spectral–spatial center attention (SSCA) module to extract spatial information across multiple scales. Wang et al. [50] introduced two self-similarity descriptors to enhance the spatial feature representations within patches, then employed a spectral interactive transformer (SIT) module and a spatial conv-attention (SCA) module to extract discriminative information. Chen et al. [51] proposed a novel spectral and spatial consistency constraint (S2C2) loss, which adaptively integrates spatial and spectral information without introducing additional parameters. However, although these methods can effectively exploit short- and long-range dependencies inherent in HSIs, they fail to consider the mid-range spatial dependencies, leading to suboptimal classification performance.

III. Methodology

In this section, we introduce the proposed MCMTN method in detail. It is mainly composed of three stages: initial feature extraction, triple-range feature extraction, and classification.

A. Initial Feature Extraction

The overall framework of our proposed MCMTN method is illustrated in Fig. 1. In the first stage, a single convolutional layer is employed to extract shallow features from HSIs and perform initial information aggregation, which is then passed to the second stage. Mathematically, the feature extraction in the first stage can be represented as follows:

$$X_{pri} = GeLU\left(BN\left(Conv_{3\times 3}\left(X_{\text{in}}\right)\right)\right) \tag{1}$$

where $X_{in} \in \mathbb{R}^{P\times P\times S}$ and X_{pri} denote the input features and the output features in the first stage, respectively. P represent the input patch size and S is the number of bands. After applying the 3×3 convolution, both batch normalization (BN) and GeLU nonlinear activation operations are subsequently performed to facilitate a more stable and efficient training process.

B. Triple-Range Feature Extraction

In the triple-range feature extraction stage, our proposed MCMTN method primarily extracts features at three different spatial scales, promoting effective interaction among spatial features. Specifically, this stage comprises two SSCM modules and one MCM module. The SSCM is designed to efficiently extract both local and global information from HSIs in the spectral and spatial domains, while the MCM module focuses on capturing mid-range spatial dependencies and facilitating semantic feature interaction. The detailed feature flow of the triple-range feature extraction stage is illustrated as follows:

$$X_{tri} = MCM\left(SSCM\left(SSCM\left(X_{pri}\right)\right)\right) \tag{2}$$

where X_{tri} represents the output features from the triple-range feature extraction stage. The SSCM and MCM modules will be introduced in detail as follows.

1) Spatial–Spectral Compression Module: Inspired by EfficientFormerV2 [52], we design the SSCM to extract spatial–spectral features from HSIs. As shown in Fig. 1, SSCM comprises an LWSSM module and an SCM module. The LWSSM is employed to extract spectral and short-range spatial features, while the SCM is responsible for capturing long-range spatial features. Furthermore, SSCM incorporates a residual structure to enhance information flow within the network. The complete process of SSCM can be formulated as follows:

$$X_{SSCM} = I\left(X_{in}\right) + SCM\left(LWSSM\left(X_{in}\right)\right) \tag{3}$$

where $I(.)$ denotes the identity mapping operation, and X_{in} and X_{SSCM} represent the input and output features, respectively. We first apply LWSSM to extract spectral and local spatial information from the input feature map, which can be formulated as

$$X_{LWSSM} = PB_2\left(DB_{3\times 3}\left(PB_1\left(X_{in}\right)\right)\right) \tag{4}$$

where $PB(.)$ denotes the pointwise convolution block and $DW_{3\times 3}(.)$ represents a depthwise convolution block with a kernel size of 3×3. The pointwise convolution block is constructed with a sequence comprising a 1×1 convolution, a BN layer, and a ReLU function. The depthwise convolution block consists only of a 3×3 convolution and a BN layer. The first pointwise convolution block $PB_1(.)$ extracts spectral information from the input feature map. BN within the sequence normalizes the input distribution, ensuring more stable nonlinear activation through ReLU. Then, $DB_{3\times 3}(.)$ performs convolution on each channel of the input feature map separately, thus capturing local spatial features within each channel. The second pointwise convolution block $PB_2(.)$ integrates the extracted spatial–spectral features and further refines the spectral information of the input feature map. Next, the SCM module extracts global spatial features from the output of the LWSSM module. As shown in Fig. 2, the SCM module first compresses the spatial–spectral features, extracts global spatial features, and then decompresses the features to generate its output. Specifically, the input features $X_{LWSSM} \in \mathbb{R}^{P\times P\times C}$ is compressed into $X^{'} \in \mathbb{R}^{C\times (P^2)}$, where $P \times P$ and C represent the spatial dimensions and the number of channels, respectively. A pointwise convolution block is used to integrate channel information in $X^{'}$, which represents the spatial features in the original feature map. Finally, after feature extraction, the features are decompressed back to $X^{out} \in \mathbb{R}^{C\times P\times P}$. The detailed computational process of the SCM module is given as follows:

$$X_{SCM} = DeZip\left(PB_3\left(Zip\left(X_{LWSSM}\right)\right)\right) \tag{5}$$

where Zip (.) and DeZip (.) denote the compression and decompression operations, respectively.

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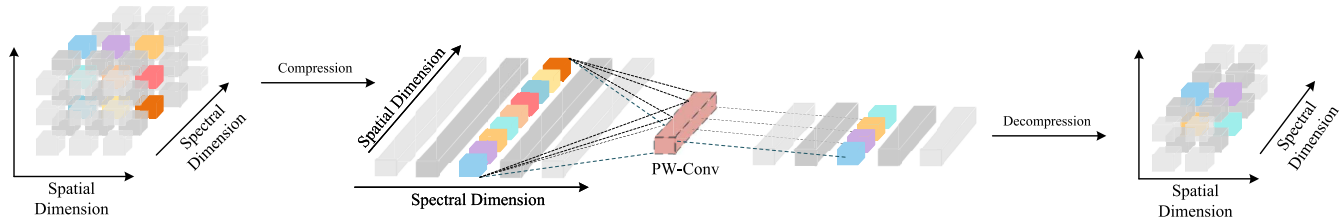


Fig. 2. Detail structure of the SCM.

2) Mid-Range Convolutional Modulation Module: To capture mid-range spatial dependencies within patches, a simple yet effective convolutional modulation layer is designed to replace the SA in the traditional transformer. In the original SA module, the input feature map is first linearly projected by three randomly initialized learnable weight matrices W^Q, W^K, and W^V. Then, we have

$$\begin{aligned} Q_{SA} &= X\left(W^Q\right)^{\mathrm{T}} \\ K_{SA} &= X\left(W^K\right)^{\mathrm{T}} \\ V_{SA} &= X\left(W^V\right)^{\mathrm{T}} \end{aligned} \tag{6}$$

where $Q, K, V \in \mathbb{R}^{P^2 \times C}$ are the input features. Then the query matrix Q_{SA} and the key matrix K_{SA} are matched for similarity by the dot product operation. Here, the gradient disappearance can be mitigated by dividing the product by $\sqrt{d_k}$, where $d_k = S$ is the dimension of pixel tokens. Then the similarity matrix is obtained by the softmax function for normalization. Finally, the similarity matrix and the value matrix V_{SA} are multiplied to get the output of SA. Mathematically, the SA can be described as follows:

$$\begin{aligned} A_{SA} &= Softmax\left(\frac{Q_{SA}K_{SA}^{\mathrm{T}}}{\sqrt{d_k}}\right) \\ X_{SA} &= A_{SA}V_{SA}. \end{aligned} \tag{7}$$

In the proposed MCM module, instead of computing a similarity matrix A_{SA} as in (7), mid-range convolutional features are used as the attention matrix. As illustrated in Fig. 3(b), the MCM module employs depthwise convolution with a large kernel and Hadamard product to compute the output features. Specifically, we employ two independent PW-Conv layers to project the input feature map. Then, the MCM module generates A_{MCM} using depthwise convolution with a large kernel. This process can be formulated as follows:

$$\begin{aligned} F &= GeLU\left(PW\left(LN\left(X_{in}\right)\right)\right) \\ V_{MCM} &= PW\left(X_{in}\right) \\ A_{MCM} &= DW_L\left(F\right) \end{aligned} \tag{8}$$

where PW denotes pointwise convolution, and DW_L is a depthwise convolution with a large kernel size. GeLU $(\cdot)$ denotes the Gaussian error linear unit activation function. LN$(.)$ represents a layer normalization operation [53] that accelerates the convergence rate of the model and enhances its generalization capabilities, simultaneously. After obtaining the large-kernel convolution features A_{MCM}, we use them to modulate V_{MCM} as follows:

$$X_{MCM} = PW\left(A_{MCM} \odot V_{MCM} + X\right) \tag{9}$$

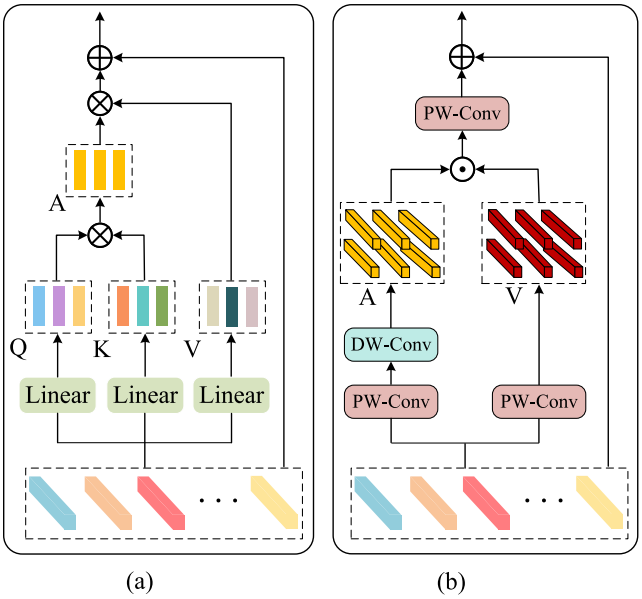


Fig. 3. Detail structure of the (a) SA and the proposed (b) MCM.

where X_{MCM} represents the output of the MCM module. In this way, we utilize the extracted mid-range spatial features more efficiently, ensuring that each pixel is correlated with pixels within a $k \times k$ neighborhood, where k is the kernel size of the large-kernel convolution. Finally, we complete the inter-channel information interaction through the pointwise convolution operation in (9). Compared to SA, the MCM module establishes spatial relationships through convolution operations, which are more memory-efficient. It can also adapt input features of varying spatial dimensions through the modulation mechanism.

C. Classification

As shown in Fig. 1, the classifier consists of a global average pooling and a fully connected layer. Finally, the spatial–spectral features are fed into the classifier to obtain the land cover categories of the pixels. Specifically, the global average pooling operation aggregates the spatial–spectral representations into a compact feature vector, which reduces the spatial dimensionality while preserving the most discriminative information. This pooling strategy enhances the generalization capability of the model by focusing on the dominant spectral–spatial characteristics. The resulting feature vectors are then fed into the fully connected layer, which maps the aggregated features into the label space and obtains the

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TABLE I
SAMPLE SIZE OF THE IP DATASET

No.	Class	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing
		A		B		C		D		E		F	
1	Alfalfa	10	36	15	31	18	28	25	21	25	21	14	32
2	Corn-notill	35	1393	100	1328	200	1228	400	1028	500	928	624	804
3	Corn-mintill	35	795	70	760	100	730	200	630	280	550	365	465
4	Corn	35	202	50	187	75	162	75	162	75	162	81	156
5	Grass-pasture	35	448	50	433	75	408	150	333	175	308	204	279
6	Grass-trees	35	695	50	680	75	655	200	530	250	480	327	403
7	Grass-pasture-mowed	10	18	12	16	12	16	15	13	15	13	15	13
8	Hay-windrowed	35	443	50	428	75	403	150	328	175	303	200	278
9	Oats	10	10	12	8	12	8	12	8	12	8	12	8
10	Soybean-notill	35	937	70	902	100	872	300	672	380	592	436	536
11	Soybean-mintill	35	2420	100	2355	200	2255	600	1855	800	1655	1127	1328
12	Soybean-clean	35	558	50	543	75	518	180	413	180	413	209	384
13	Wheat	35	170	50	155	50	155	60	145	70	135	80	125
14	Woods	35	1230	100	1165	200	1065	350	915	500	765	1032	233
15	Building-Grass-Trees-Drives	35	351	50	336	75	311	120	266	135	251	150	236
16	Stone-Steel-Towers	10	83	15	78	25	68	25	68	30	63	34	59
Total		460	9789	844	9405	1367	8882	2862	7387	3602	6647	4910	5339

classification results. In our MCMTN method, cross-entropy [54] is used to calculate the loss between the predicted results and the ground truth. The loss function of the proposed MCMTN method is given as follows:

$$\text{Loss}_{CE} = \frac{1}{N}\sum_{n=1}^{N}\text{Loss}_i = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} y_c^n \log\left(o_c^n\right) \tag{10}$$

where y_c^n is the ground truth label and o_c^n denotes the predictive label. C is the number of land cover categories, and N is the sample number in a minibatch.

IV. Experiment and Analysis

In this section, we conduct experiments on three widely used benchmark datasets to evaluate the effectiveness of the proposed MCMTN method. Several state-of-the-art HSI classification methods are employed for comparison under various evaluation metrics.

A. Data Description

1) Indian Pines: The Indian Pines (IP) dataset was gathered by the AVIRIS sensor over the Indian Pines test site in North-western Indiana, which consists of 145 × 145 pixels. The dataset contains 16 ground truth classes, which are not mutually exclusive. After removing the bands corresponding to water absorption regions, the number of spectral bands was reduced to 200.

2) Houston 2013: The Houston 2013 (HUST2013) dataset was collected by the ITRES CASI-1500 sensor in June 2012 over the University of Houston campus and adjacent urban areas in Texas, which was provided by the IEEE GRSS Data Fusion Competition 2013 [55]. Houston 2013 has 144 spectral bands with a spatial resolution of 349 × 1905 pixels. In addition, there are 15 land-cover classes in the HUST2013 dataset.

3) WHU-Hi-Longkou: The Longkou (LK) dataset was captured in 2018 in Longkou Town, Hubei Province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. The dataset encompasses nine land cover types.

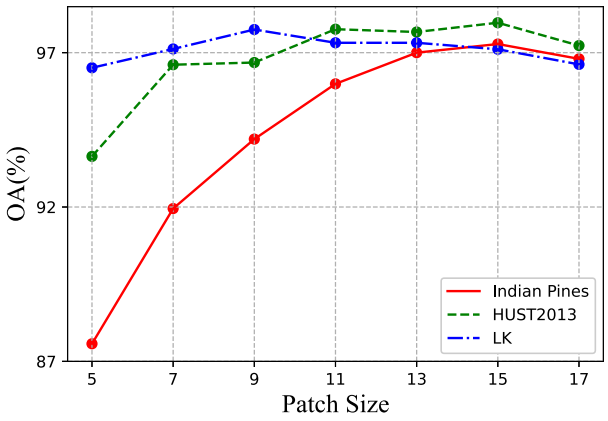


Fig. 4. OA of the proposed MCMTN method with different patch sizes.

The UAV captured imagery at an altitude of 500 m, with a spatial resolution of approximately 0.463 m and an image size of 550 × 400 pixels. Its spatial resolution is approximately 0.463 m.

B. Experimental Settings and Evaluation Metrics

To ensure a fair evaluation of the proposed method, all unlabelled samples were excluded, and the remaining labeled samples were randomly divided into training and testing sets. Tables I–III show the sample sizes of the training and testing sets for all HSI datasets. It is worth noting that we referred to their amounts using the corresponding code names in the tables. In this study, performance evaluation experiments were conducted on the data partition labeled as Code Name A. The other partitioning strategies were utilized in Section IV-G.

The hardware for the experiment was configured as follows: 13th Gen Intel[1] Core[2] i7-13700H CPU, NVIDIA GeForce RTX 3090 GPU with 24-GB memory. In addition, the software configuration included Windows 11, Python 3.9.21, PyTorch 2.3.1, and CUDA 12.1.

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TABLE II
SAMPLE SIZE OF THE HUST2013 DATASET

No.	Class	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing
		A		B		C		D		E		F	
1	Healthy grass	50	1201	100	1151	200	1051	300	951	400	851	585	666
2	Stressed grass	50	1204	100	1154	200	1054	300	954	400	854	587	667
3	Synthetic grass	10	687	20	677	40	657	60	637	80	617	338	359
4	Tree	50	1194	100	1144	200	1044	300	944	400	844	582	662
5	Soil	50	1192	100	1142	200	1042	300	942	400	842	586	656
6	Water	10	315	20	305	40	285	60	265	80	245	154	171
7	Residential	50	1218	100	1168	200	1068	300	968	400	868	594	674
8	Commercial	50	1194	100	1144	200	1044	300	944	400	844	582	662
9	Road	50	1202	100	1152	200	1052	300	952	400	852	587	665
10	10Highway	50	1177	100	1127	200	1027	300	927	400	827	573	654
11	Railway	50	1185	100	1135	200	1035	300	935	400	835	577	658
12	Parking lot1	50	1183	100	1133	200	1033	300	933	400	833	576	657
13	Parking lot2	10	459	20	449	40	429	60	409	80	389	224	245
14	Tennis court	10	418	20	408	40	388	60	368	80	348	204	224
15	Running track	10	650	20	640	40	620	60	600	80	580	320	340
Total		550	14479	1100	13929	2200	12829	3300	11729	4400	10629	7069	7960

TABLE III
SAMPLE SIZE OF THE LK DATASET

No.	Class	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing	Trainning	Testing
		A		B		C		D		E		F	
1	Corn	34	34476	172	34338	345	34165	1725	32785	3451	31059	6902	27608
2	Cotton	8	8365	41	8332	83	8290	418	7955	837	7536	1674	6699
3	Sesame	3	3027	15	3015	30	3000	151	2879	303	2727	606	2424
4	Broad-leaf soybean	63	63148	316	62895	632	62579	3160	60051	6321	56890	12642	50569
5	Narrow-leaf soybean	4	4146	20	4130	41	4109	207	3943	415	3735	830	3320
6	Rice	11	11842	59	11794	118	11735	592	11261	1185	10668	2370	9483
7	Water	67	66988	335	66720	670	66385	3352	63703	6705	60350	13411	53644
8	Roads and houses	7	7116	35	7088	71	7052	356	6767	712	6411	1424	5699
9	Mixed weed	5	5223	26	5202	52	5176	261	4967	522	4706	1045	4183
Total		202	204331	1019	203514	2042	202491	10222	194311	20451	184082	40904	163629

Although larger input patch sizes contain richer spectral–spatial information, they also introduce more noise. Therefore, we empirically determined the optimal patch size for each dataset. As illustrated in Fig. 4, the patch size was set to 15 × 15 for the IP and HUST2013 datasets, and 9 × 9 for the LK dataset. The training batch size was consistently set to 64 across all three datasets. To accelerate model convergence, a momentum-based optimizer was adopted in conjunction with a learning rate decay strategy. Specifically, the momentum and weight decay parameters were empirically set to 0.8 and 0.0001, respectively.

To quantitatively analyze the effectiveness of the proposed method, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (KAPPA) were used to evaluate the classification results. In general, the closer the values of these three indicators are to 1, the better the classification performance and the more accurate the results. All experiments were conducted ten times under the same conditions, and the model parameters corresponding to the highest OA were selected as the final configuration. This optimized model was then used to predict the category of each pixel, and the resulting predictions were used to generate the corresponding feature maps.

C. Classification Performance

To verify the superiority of the proposed model, we employed several well-known deep HSI classification models, such as SSAN [56],[3] LS^{2}CM [57],[4] GAHT [45],[5] SiT [51],[6] DSFormer [58],[7] MorphFormer [41],[8] LSAViT [48],[9] SSFTT [46],[10], and MASSFormer [47][11] as the comparison methods. These comparison methods not only include classic and state-of-the-art CNN models, but also contain some transformer-based HSI classification models.

In this subsection, we quantitatively analyzed the classification performance of all HSI classification methods in terms of classification accuracy and visualization feature maps. For a fair comparison, all models were trained and evaluated using the same training and testing sets. Tables IV–VI show the classification results of our MCMTN model and other comparison models on the IP, HUST2013, and LK datasets. Figs. 5–7 plot the feature maps of different models on the IP, HUST2013, and LK datasets. From the experimental results on different benchmark datasets, we can draw the following conclusions.

1) It can be seen that the proposed MCMTN model achieves the best classification accuracy on all datasets.

[3]https://github.com/EtPan/SSAN
[4]https://github.com/zhe-meng/LSSCM
[5]https://github.com/MeiShaohui/Group-Aware-Hierarchical-Transformer
[6]https://github.com/zhonghaochen/SiT
[7]https://github.com/YichuXu/DSFormer
[8]https://github.com/mhaut/morphFormer
[9]https://github.com/machao132/LSGA-VIT
[10]https://github.com/zgr6010/HSI_SSFTT
[11]https://github.com/hz63/MASSFormer

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE IP DATASET

No.	SSAN	LS^{2}CM	GAHT	SiT	DSFormer	MorphFormer	LSAViT	SSFTT	MASSFormer	**MCMTN**
1	90.91	79.49	77.50	83.78	76.92	78.95	83.78	77.50	67.39	86.11
2	91.02	93.55	85.20	88.52	92.57	90.96	94.59	94.70	86.82	96.27
3	90.63	95.60	95.66	95.77	91.04	92.66	97.82	96.36	83.14	97.11
4	87.74	93.50	87.26	93.4	92.08	95.90	96.39	92.54	85.71	95.36
5	90.12	97.39	94.42	97.41	93.84	95.24	97.15	97.02	96.47	98.62
6	96.87	96.41	97.17	98.52	97.5	99.27	96.79	97.67	98.81	99.41
7	76.47	92.86	100.00	72.22	86.67	100.00	92.86	100.00	92.86	100.00
8	98.13	99.77	98.84	98.6	99.06	99.07	99.77	99.53	99.53	100.00
9	7.04	41.67	15.15	27.78	21.74	38.46	17.86	31.25	31.25	28.57
10	76.33	95.84	89.59	90.24	80.66	92.80	95.72	96.18	75.46	96.26
11	94.26	98.12	95.46	98.23	98.74	99.00	97.61	97.61	96.05	98.29
12	72.19	92.55	80.37	89.44	95.76	90.47	91.94	97.97	84.44	93.84
13	95.59	95.65	94.51	98.09	98.04	96.82	92.64	94.48	95.68	99.35
14	98.75	99.09	97.71	98.03	95.37	98.44	98.68	99.34	99.17	99.67
15	85.48	95.71	94.38	95.93	97.04	96.24	92.05	96.25	94.63	97.10
16	76.24	91.57	86.21	86.05	80.00	87.21	88.10	90.70	78.00	90.48
OA(%)	89.30	96.32	92.19	94.67	93.67	95.47	96.13	96.79	90.45	**97.49**
AA(%)	82.98	91.17	86.84	88.25	87.31	90.72	89.61	91.19	85.34	**92.43**
KAPPA×100	87.80	95.78	91.06	93.9	92.76	94.82	95.56	96.32	89.13	**97.12**

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE HUST2013 DATASET

No.	SSAN	LS^{2}CM	GAHT	SiT	DSFormer	MorphFormer	LSAViT	SSFTT	MASSFormer	**MCMTN**
1	96.93	95.72	96.27	93.09	96.40	96.17	92.00	97.56	95.82	98.83
2	98.33	99.74	98.73	99.38	98.33	98.50	94.62	98.59	99.66	99.75
3	99.85	100.00	99.25	99.70	99.25	99.25	99.85	98.23	98.37	99.55
4	97.71	98.03	97.66	95.84	98.27	97.66	96.50	96.52	98.53	99.91
5	95.46	99.58	97.52	96.80	96.02	97.85	99.49	97.52	95.55	98.50
6	97.30	100.00	99.31	98.66	97.36	98.32	98.95	94.23	92.16	100.00
7	95.02	92.60	89.46	90.98	92.82	96.45	90.04	95.01	89.74	95.17
8	95.35	99.47	98.41	99.38	99.36	99.82	100.00	99.82	98.56	99.48
9	91.44	93.48	92.48	95.65	93.78	90.81	94.83	94.53	96.72	95.60
10	96.32	100.00	98.48	95.42	98.97	97.39	98.23	98.72	97.63	99.40
11	94.75	99.56	99.39	97.53	99.04	97.06	99.57	99.04	99.38	99.13
12	97.86	99.57	98.49	99.65	97.55	100.00	98.64	98.61	98.64	98.54
13	87.16	97.32	94.07	94.39	91.45	98.60	96.77	92.18	91.45	96.64
14	97.71	100.00	100.00	99.5	99.50	100.00	100.00	100.00	100.00	100.00
15	93.59	97.67	98.11	97.66	96.74	95.85	97.06	97.80	96.16	97.20
OA(%)	95.75	97.91	96.85	96.58	96.98	97.29	96.67	97.41	96.73	**98.43**
AA(%)	95.65	98.18	97.18	96.91	96.99	97.58	97.10	97.22	96.56	**98.52**
KAPPA×100	95.4	97.74	96.60	96.30	96.74	97.07	96.40	97.21	96.46	**98.3**

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT HSI CLASSIFICATION MODELS ON THE LK DATASET

No.	SSAN	LS^{2}CM	GAHT	SiT	DSFormer	MorphFormer	LSAViT	SSFTT	MASSFormer	**MCMTN**
1	98.05	97.92	99.32	98.85	94.22	98.35	99.33	96.82	97.34	99.54
2	91.48	95.43	97.94	93.46	87.20	86.80	68.56	97.10	84.85	98.37
3	89.62	94.61	90.71	96.72	81.62	93.63	95.70	98.31	90.00	99.21
4	92.53	94.69	96.62	96.66	96.85	92.13	94.43	96.19	92.38	95.42
5	67.49	98.99	88.65	93.51	81.48	80.80	95.09	94.02	73.22	93.40
6	98.80	90.23	99.78	99.84	95.44	97.69	99.74	98.29	91.52	99.74
7	98.82	99.89	98.42	99.13	98.25	99.83	99.22	97.92	99.79	99.67
8	79.32	81.19	87.37	88.04	85.47	77.68	88.45	91.31	89.29	85.69
9	84.83	99.08	96.91	98.63	86.92	93.07	99.51	88.33	70.93	99.06
OA(%)	94.65	96.23	97.35	97.35	95.30	95.18	95.62	96.73	94.62	**97.57**
AA(%)	88.99	94.67	95.08	96.09	89.72	91.11	93.34	95.36	87.70	**96.68**
KAPPA×100	92.90	95.01	96.51	96.57	93.80	93.60	94.20	95.67	92.88	**96.78**

As shown in Table IV, on the IP dataset, our proposed MCMTN achieves the highest OA value (97.49%), AA, and KAPPA value (97.12%) among all HSI classification models. Specifically, the MCMTN model improves

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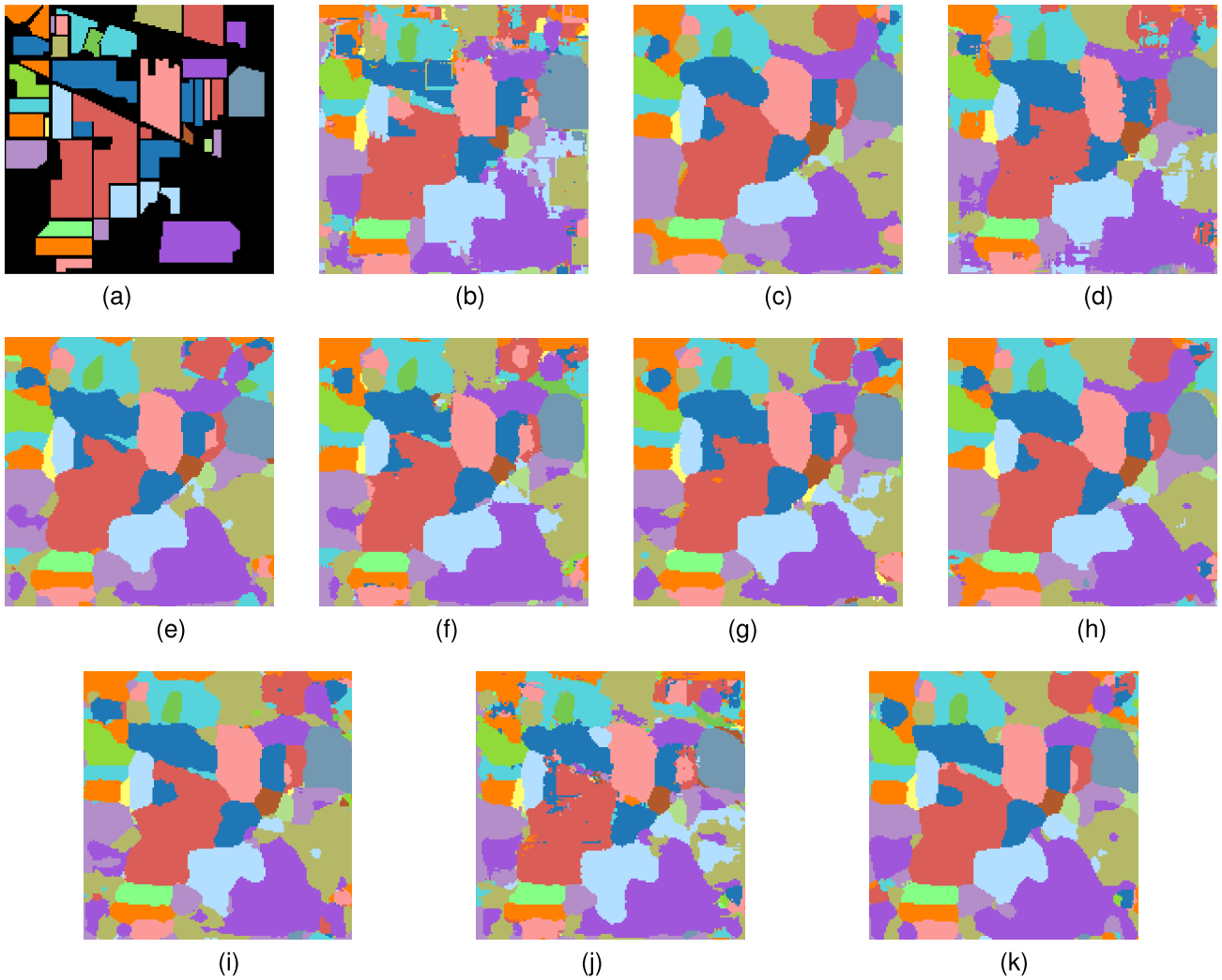


Fig. 5. Classification maps of different models on the IP dataset. (a) Ground truth. (b) SSAN. (c) LS^{2}CM. (d) GAHT. (e) SiT. (f) DSFormer. (g) morphFormer. (h) LSAViT. (i) SSFTT. (j) MASSFormer. (k) MCMTN.

the OA value by 8.19% (SSAN), 1.17% (LS^{2}CM), 5.30% (GAHT), 2.82%(SiT), 3.82% (DSFormer), 2.02% (MorphFormer), 1.36% (LSAViT), 0.70% (SSFTT), and 7.04% (MASSFormer), respectively. Table V displays the performances of different models on the HUST2013 dataset. It is easy to see that our MCMTN model achieves the highest OA value (98.43%), AA value (98.52%), and KAPPA value (98.30%). The improvement of OA values obtained by the proposed MCMTN model is 2.68% (SSAN), 0.52% (LS^{2}CM), 1.58% (GAHT), 1.85%(SiT), 1.45% (DSFormer), 1.14% (MorphFormer), 1.76% (LSAViT), 1.02% (SSFTT), and 1.70% (MASSFormer), respectively. Table VI shows the classification results of different methods on the LK dataset. Our MCMTN model also obtains the highest OA value (97.57%), AA value (96.68%), and KAPPA value (96.78%) on the LK dataset. In terms of OA metric, our proposed MCMTN model has increased by 2.92% (SSAN), 1.34% (LS^{2}CM), 0.22% (GAHT), 0.22%(SiT), 2.27% (DSFormer), 2.39% (MorphFormer), 1.95% (LSAViT), 0.84% (SSFTT), and 2.95% (MASSFormer), respectively.

2) As observed in Figs. 5–7, the feature maps of our MCMTN method on different datasets are the closest to ground truth among all methods, which is consistent with the conclusions presented in Tables IV–VI. The classification maps produced by MASSFormer contain noticeable salt-and-pepper noise, which can be attributed to its limited ability to exploit local information. Generally, most methods struggle with boundary pixel classification, as the patches centered on boundary pixels often contain a higher proportion of irrelevant pixels, thereby reducing the final classification accuracy. However, it is clear to see that the feature maps of the proposed MCMTN on different datasets include the least noisy points and the smoothest class boundaries compared with other HSI classification models. It further demonstrates the superiority of our proposed MCMTN method in HSI classification problems.

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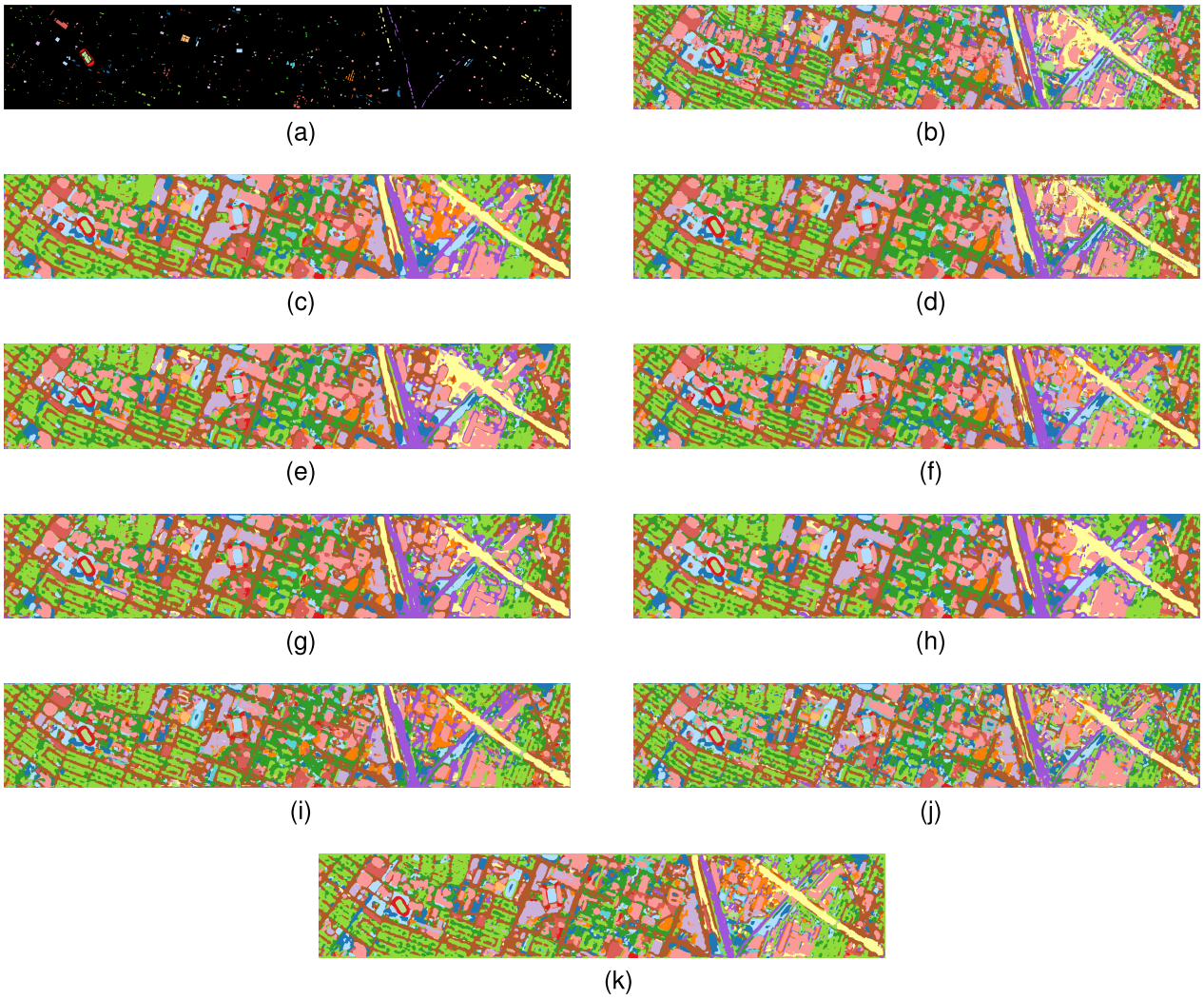


Fig. 6. Classification maps of different models on the HUST2013 dataset. (a) Ground truth. (b) SSAN. (c) LS^{2}CM. (d) GAHT. (e) SiT. (f) DSFormer. (g) morphFormer. (h) LSAViT. (i) SSFTT. (j) MASSFormer. (k) MCMTN.

3) On all three datasets, CNN-based HSI classification models (i.e., SSAN and LS^{2}CM) can achieve relatively promising classification performance. This is because they fully leverage the local spatial–spectral features of HSIs. However, SSAN exhibits the weakest performance across all three datasets. This can be attributed to its relatively simple architecture, which fails to capture sufficiently discriminative features. Although other transformer-based HSI classification models (i.e., GAHT, SiT, DSFormer, MorphFormer, LSAViT, SSFTT, and MASSFormer) further capture long-distance dependencies, they cannot consistently outperform pure convolution approaches. This indicates that combining CNNs and transformers is beneficial for classification, but their respective strengths must be carefully balanced. In contrast, the proposed MCMTN achieves the best classification performance across various datasets, as it efficiently captures both global and local information while dynamically extracting mid-range features, thereby fully exploiting spatial information within patches.

TABLE VII

PARAMS, FLOPS, TRAINING TIME, AND INFERENCE TIME OF TRANSFORMER-BASED HSIC METHODS ON THE IP DATASET

Method	Params(M)	FLOPs(G)	Train(s)	Infer(s)
GAHT	1.0381	14.94	64.2541	2.3504
SiT	1.8378	22.82	185.7313	6.4038
DSFormer	0.7897	5.43	291.7209	11.8585
MorphFormer	**0.1985**	4.18	109.8976	4.2850
LSAViT	1.1575	17.23	85.7954	3.4709
SSFTT	0.9318	7.60	33.3655	1.8322
MASSFormer	1.0876	8.92	42.2961	1.6351
Ours	0.2605	**2.89**	**30.7030**	**1.334**

4) Table VII presents the complexity of all transformer-based methods on the IP dataset, evaluated in terms of the number of parameters (Params), floating point operations (FLOPs), training time (Train), and inference time (Infer). To ensure fairness, all methods were tested

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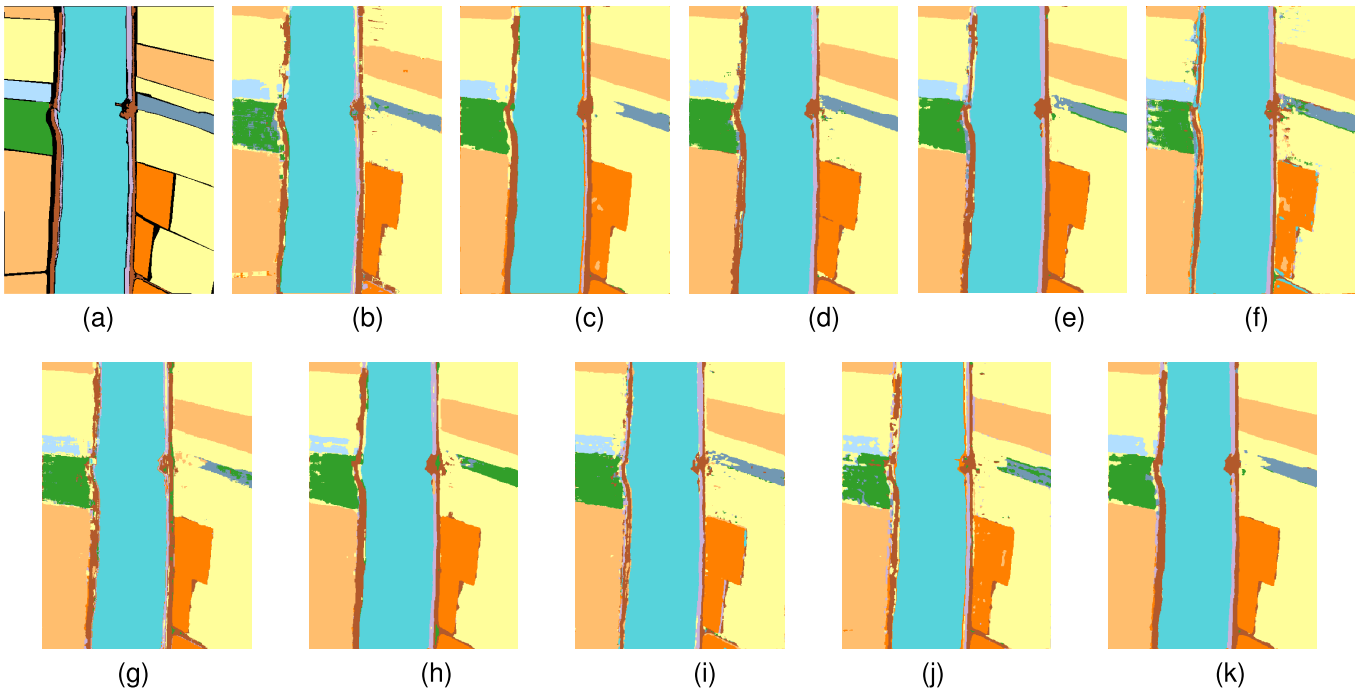


Fig. 7. Classification maps of different models on the LK dataset. (a) Ground truth. (b) SSAN. (c) LS^{2}CM. (d) GAHT. (e) SiT. (f) DSFormer. (g) morphFormer. (h) LSAViT. (i) SSFTT. (j) MASSFormer. (k) MCMTN.

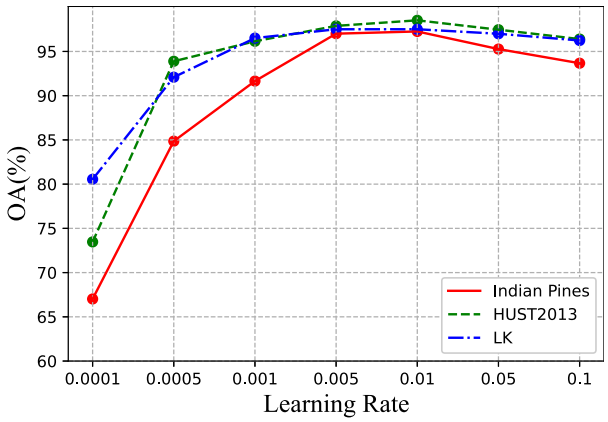


Fig. 8. OA of our proposed MCMTN method with different learning rates.

on the same computational platform. Factors such as preprocessing time and accuracy calculation time were excluded, and only the pure training and inference times were recorded. Notably, except for the MorphFormer, our proposed MCMTN method has fewer parameters than all other competitors while still achieving the highest classification accuracy. In addition, the results show that the proposed method offers clear advantages in both training and inference speed, indicating more efficient information extraction and a distinct benefit for real-time applications.

D. Influence of the Learning Rate

In this subsection, we evaluated the influence of different learning rate settings in the proposed model. The learning rate determines the convergence during the training process. It is noteworthy that the model may skip over the optimal point by setting a large learning rate. Meanwhile, the model may fail to reach the optimal point and fall into a locally optimal point by choosing a small learning rate. Therefore, the learning rate is closely related to the final performance of the model. Here, we empirically set several commonly used learning rate values, and qualitatively compared different learning rate settings on IP, HUST2013, and LK, respectively. Then the optimal learning rate was chosen to train the proposed model. The experimental results with different learning rate settings on three benchmark datasets are shown in Fig. 8. It is observed that a learning rate of 0.01 yields the highest classification accuracy on the IP (97.49%), HUST2013 (98.43%), and LK (97.57%) datasets. Therefore, we set the learning rate to 0.01 for all further experiments with our proposed model.

E. Influence of the Epochs

In this part, we conducted some experiments to evaluate the performance of our proposed MCMTN method with different training epoch settings. Fig. 9 shows the classification performances of our MCMTN method with different training epochs on three datasets. It can be found that we trained a total of 21 model weights from 25 epochs to 300 epochs on the IP, HUST2013, and LK datasets. We can see that MCMTN can only achieve OA of 89.75%, 94.67%, and 92.15% on IP, HUST2013, and LK datasets with 25 training epochs, respectively. The proposed MCMTN converges faster with the increase of training epochs. On the HUST2013 and LK datasets, the highest OA values of 98.43% and 97.57% are achieved after 200 training epochs, respectively. On the IP dataset, the highest OA of 97.49% is achieved at 150

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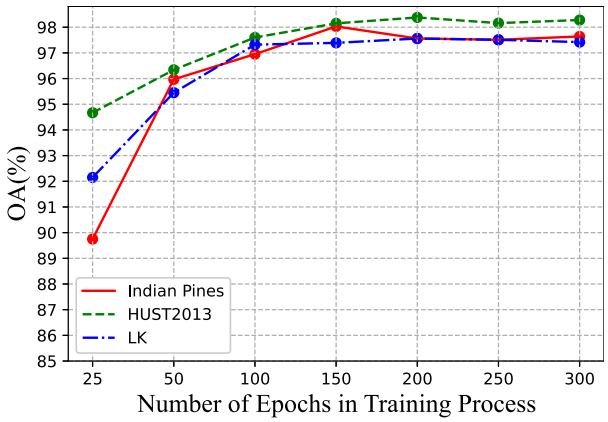


Fig. 9. OA of our MCMTN method with different training epochs.

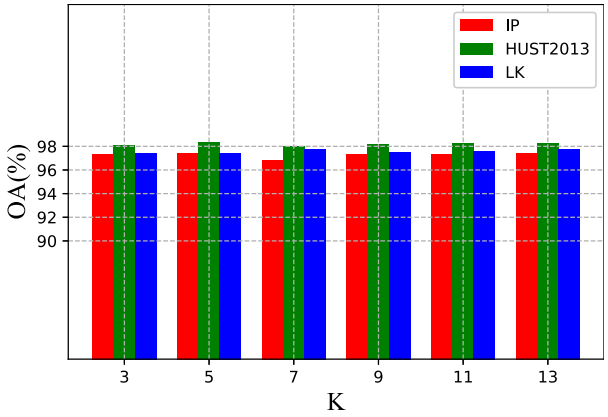


Fig. 10. OA of our MCMTN method with different large convolution kernel sizes in MCM.

training epochs. However, the classification performance of MCMTN degrades with more than 200 training epochs. This phenomenon may be attributed to overfitting in the proposed MCMTN model. Therefore, in our experiments, we set the number of training epochs to 150 for the IP dataset and 200 for the HUST2013 and LK datasets.

F. Influence of Large Convolution Kernel Size in MCM

To explore the adaptability of the proposed MCMTN method on different datasets, we investigated the impact of large convolution kernel sizes in MCM on the classification performance. In MCMTN, the kernel size of the DW-Conv is set from 3×3 to 13×13. The classification results of the proposed MCMTN method with different large convolution kernel sizes are shown in Fig. 10. When the selected kernel sizes (K) were configured as 5, 5, and 7, the highest OA achieved on the IP, HUST2013, and LK datasets were 97.49%, 98.43%, and 97.57%, respectively. These kernel sizes were also used in all subsequent experiments involving our MCMTN method.

G. Influence of Training Set Size

To validate the generalization performance of the proposed MCMTN method, we conducted some experiments to analyze the impact of different training sample sizes on classification

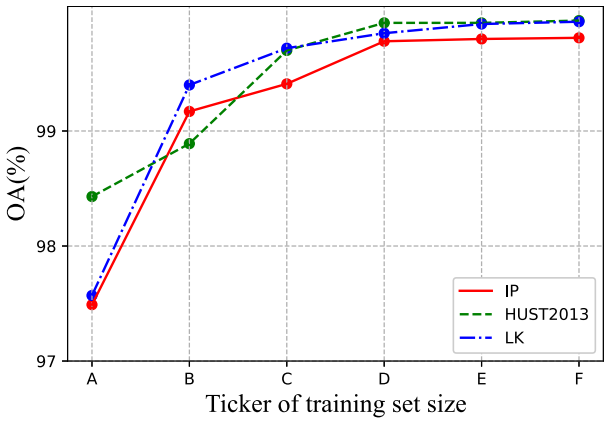


Fig. 11. OA of our proposed MCMTN method with different training sample sizes.

performance. Fig. 11 shows the results of the proposed model with different training sample sizes. Here, we randomly chose different numbers of HSI samples from IP, HUST2013, and LK datasets as training sets. Tables I–III show the numbers of the training and testing samples on three datasets. We can see that the performance of the proposed method can be improved with the increase of the training sample size on different datasets. However, the performance increment of the model decreases as the training set size continues to increase on IP, HUST2013, and LK datasets. This is because the proposed MCMTN model keeps relatively stable performances on the three datasets once it has been trained with sufficient training samples. This phenomenon is consistent with our understanding that the performance of deep learning models is closely related to the size of the training dataset.

H. Ablation Studies

To fully demonstrate the effectiveness of the components in the proposed MCMTN method, we carried out ablation experiments on the IP, HUST2013, and LK datasets. To measure the impact of each component on the classification performance, three variants of the proposed MCMTN method were constructed in the ablation experiments. Specifically, for case 1, we removed the LWSSM of the model to verify its effectiveness. For case 2, we removed the SCM block, thereby omitting the compression convolution operation. For case 3, we removed the MCM module from the model to demonstrate its superiority in the MCMTN model. The last case is the original MCMTN model, which uses all components for HSI classification tasks. The experimental results on the three datasets are shown in Table VIII, with the highest value in each column highlighted in bold. Compared to the original MCMTN model, removing the LWSSM caused a decrease in OA of 11.35%, 11.62%, and 7.61% on the IP, HUST2013, and LK datasets, respectively. In addition, removing the SCM module led to a decrease in OA of 0.7%, 1.25%, and 0.02% on the three datasets, respectively. Similarly, removing the MCM module caused a drop in OA of 0.16%, 0.17%, and 0.07% on the three datasets, respectively. These results demonstrate that the MCMTN model consistently outperforms these three

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TABLE VIII
ABLATION EXPERIMENTS OF THE PROPOSED MCMTN METHOD ON DIFFERENT DATASETS

Case	Component			IP			HUST2013			LK		
	LWSSM	SCM	MCM	OA(%)	AA(%)	KAPPA×100	OA(%)	AA(%)	KAPPA×100	OA(%)	AA(%)	KAPPA×100
1	×	✓	✓	86.14	80.11	84.26	86.81	89.68	85.74	89.96	88.94	86.49
2	✓	×	✓	96.79	91.30	96.32	97.18	97.50	96.95	97.55	96.58	96.76
3	✓	✓	×	97.33	92.36	96.93	98.26	98.47	98.13	97.50	96.22	96.73
4	✓	✓	✓	**97.49**	**92.43**	**97.12**	**98.43**	**98.52**	**98.30**	**97.57**	**96.68**	**96.78**

TABLE IX
ABLATION STUDIES OF OUR MCMTN METHOD ON THREE DATASETS

Methods	IP			HUST2013			LK		
	OA(%)	AA(%)	KAPPA×100	OA(%)	AA(%)	KAPPA×100	OA(%)	AA(%)	KAPPA×100
LWSSM → Convolution2d	88.46	82.26	86.86	90.10	91.85	89.29	93.29	88.95	91.09
SCM → Convolution2d	96.84	92.37	96.38	97.39	97.65	97.18	97.50	96.20	96.69
MCM → SA	96.65	92.35	96.16	98.33	98.47	98.19	97.29	95.92	96.42
Our MCMTN method	**97.49**	**92.43**	**97.12**	**98.43**	**98.52**	**98.30**	**97.57**	**96.68**	**96.78**

variants on different HSI datasets, confirming the effectiveness of the LWSSM, SCM, and MCM within the MCMTN architecture.

In addition to module removal experiments, we conducted a series of ablation experiments to further validate the effectiveness of the LWSSM, SCM, and MCM modules. The ablation results on the three datasets are shown in Table IX, with the highest value in each column highlighted in bold. We replaced the LWSSM and SCM modules with standard 1×1 convolutions to assess their relative contribution. As a result, the performance on all three datasets declined. Moreover, the MCM module was replaced with an SA block to evaluate the advantage of MCM over SA within the MCMTN. On the IP, HUST2013, and LK datasets, replacing MCM with SA led to OA decreases of 0.84%, 0.1%, and 0.28%, respectively. This can be attributed to the fact that the proposed MCM module provides more comprehensive multirange information compared to the original SA block. These consistent ablation results demonstrate that each component of the proposed MCMTN method contributes significantly to enhancing performance in HSIC tasks.

V. CONCLUSION

This article presents a novel deep classification model, called MCMTN, for HSI classification. The proposed MCMTN method effectively captures multirange information through the SSCM module, which integrates LWSSM and SCM components. Specifically, LWSSM efficiently extracts spectral features and short-range spatial details, while SCM captures long-range spatial representations by compressing features along the spatial dimension. In addition, the proposed MCM module simplifies the SA using large-kernel convolutions, enabling the model to focus on rich mid-range spatial dependencies within the HSIs. Experiments on several benchmark datasets have demonstrated the effectiveness of our proposed MCMTN method in HSI classification tasks.

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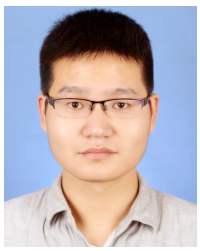
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